Biogenic amines in fish: roles in intoxication, spoilage, and nitrosamine formation--a review.

Biogenic amines are non-volatile amines formed by decarboxylation of amino acids. Although many biogenic amines have been found in fish, only histamine, cadaverine, and putrescine have been found to be significant in fish safety and quality determination. Despite a widely reported association between histamine and scombroid food poisoning, histamine alone appears to be insufficient to cause food toxicity. Putrescine and cadaverine have been suggested to potentiate histamine toxicity. With respect to spoilage on the other hand, only cadaverine has been found to be a useful index of the initial stage of fish decomposition. The relationship between biogenic amines, sensory evaluation, and trimethylamine during spoilage are influenced by bacterial composition and free amino acid content. A mesophilic bacterial count of log 6-7 cfu/g has been found to be associated with 5 mg histamine/100 g fish, the Food and Drug Administration (FDA) maximum allowable histamine level. In vitro studies have shown the involvement of cadaverine and putrescine in the formation of nitrosamines, nitrosopiperidine (NPIP), and nitrosopyrrolidine (NPYR), respectively. In addition, impure salt, high temperature, and low pH enhance nitrosamine formation, whereas pure sodium chloride inhibits their formation. Understanding the relationships between biogenic amines and their involvement in the formation of nitrosamines could explain the mechanism of scombroid poisoning and assure the safety of many fish products.